AI-POWERED SMART GARBAGE MONITORING AND CLEANING SYSTEM

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***ABSTRACT*: In this system the real VNC viewer software is used to operate the smart waste-bin by controlling from distance. When the bin becomes full, the operator starts the checking process by pressing the 'Q' key on the laptop. This operation turns on the camera, captures the images and present condition of the bin, and automatically sends a "Bin Full" signals to the selected telegram channel along with the bins location. Once the cleaning staff cleans the bin, they pushes the push button placed near the bin to confirm the work. Then the system sends a "Bin Cleaned" Message with the updated location details to telegram. If the operator presses the 'Q' key again after cleaning, the system sends a "Bin Normal" Message showing that the bin is in normal condition. At any time, if the manual push button is pressed, the system sends a "Manual Button Alert" to telegram, without considering the bin condition. This method provides continous updates and allows real-time monitoring of the smart waste-bin system.**

Index Terms: Remote Monitoring, RealVNC Viewer, Camera-Based Detection, Manual Alert System, Telegram Alerts.

I. INTRODUCTION

Municipal waste management plays a important role in maintaining public cleanliness and hygiene, but regularly monitoring dustbins is still a big challenge. A large number of bins are placed across large regions, making manual inspection difficult, time-consuming and inefficient. When waste collection depends only on manual inspection, bins may overflow before anyone notices, causing bad smell and spreading unhygienic surroundings. Due to this conditions insects and stray animals gets attracted and leading to severe health issues. Also fixed monitoring systems installed at limited locations are not always accurate. Problems such as uneven waste buildup or blocked sensors can give false readings, leading to delays in waste detection and poor cleanliness management.

Due to unnecessary manual inspections many cities have faced problems regarding dustbin overflows, poorly planned cleaning schedules and increased maintenance costs. These overflowing bins damage the environment, reduce public satisfaction, and increase the workload on waste-collection workers. So, there is a clear need for a real-time monitoring systems that reduces human effort, prevents inspection from being missed and improves overall waste management efficiency.

To solve this problems, this paper introduces an intelligent waste-bin monitoring system that integrates a Raspberry Pi, camera based detection, GPS location tracking, and real-time messaging services. Instead of depending only on manual inspections, the operator can monitor the bin remotely using the RealVNC Viewer. When the bin becomes full the operator presses the 'Q' key, which turns on the camera and captures the current condition of the bin. Then automatically send a "Bin Full" signal along with exact location and image to the municipal telegram channel. This instant alert helps authorities to send cleaning staff quickly and prevents delays in waste collection.

This system also provides a simple way to confirm that cleaning is completed. By pressing the push button fixed on the bin the cleaning staff confirms that the waste is removed. This action sends a "Bin Cleaned" message along with the updated images and location details. If needed, the operator can press the ‘Q’ key again to check the bin status and system updates it to "Bin Normal". To avoid confusion and missed status updates this confirmation process helps to complete this cycle.

Figure 1 shows that traditional manual inspection methods have caused a gradual increase in bin overflow cases over the years. On the other hand the proposed monitoring system reduces delays and improves response time by providing timely updates. By using real-time alerts, accurate location tracking, and camera-based verification, the system improves the efficiency of municipal waste management and helps keep public areas cleaner with fewer bin overflows.

In 2016, about 15 dustbin overflow cases were reported in one year. At that time, bins were checked manually, so delays in inspection often caused overflow, especially in busy public areas.

In 2017, the number of overflow cases increased to around 18. At that time the inspection process continued to depend on manual checking, which caused many bins to reamain unchecked for long periods.

In 2018, the number of dustbin overflow incidents increased to around 22. The increasing workload made it difficult for municipal workers to check all bins on time, leading to more waste being left uncollected.

In 2019, nearly 25 dustbin overflow cases were recorded. This steady increase showed that manual inspection was no longer effective for managing a growing number of dustbins in different locations.

In 2020, almost 30 dustbin overflow incidents were reported. Due the increase time gap between inspection and cleaning, causing many bins to overflow before action was taken.

In 2021, the number of dustbin overflow incidents increased to almost 35. The manual monitoring system faced difficulties due to the rapid increase in waste generation and large delays in waste collection increased due to absence of real-time monitoring.

In 2022, the number of dustbij overflow incidents increased to almost 38. This shows that manual checking became less effective, as workers could not visit all dustbin locations regularly, leading to more waste accumulation.

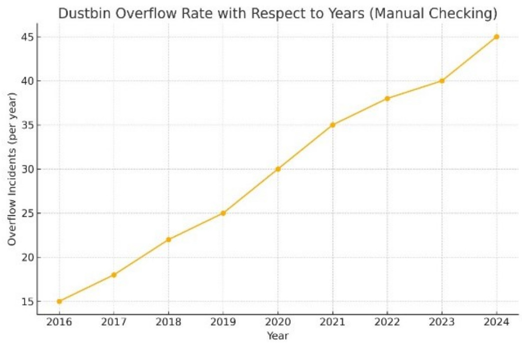

Fig.1Annual Analysis of Dustbin Overflow Incidents Using Manual Checking

In 2023, almost 40 dustbin overflow incidents taken place. This increase clearly explains that technological assistance is very effective for waste management due to increase in depending on only manual inspection methods.

In 2024, almost 45 dustbin incidents taken place, highest recorded in the mean time. This trend shows that as the number of bins and waste increased, manual monitoring became completely inadequate, resulting in frequent overflows and poor waste management.

II. Literature Survey

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III. Proposed Method

A. Procedure For Inspection

Fig. 2 Block diagram of the monitoring process. The proposed method for monitoring the AI-powered smart garbage monitoring and cleaning system is based entirely on image processing, artificial intelligence, and GPS-enabled notifications, eliminating the need for traditional sensors.

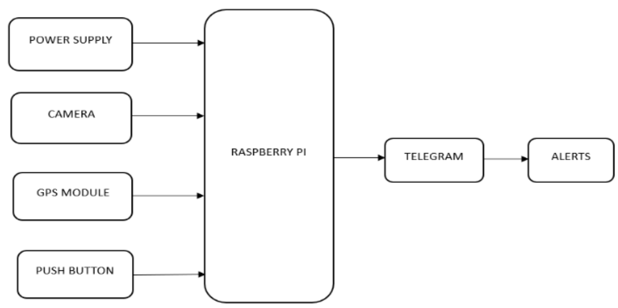

Fig.2. Block Diagram of Smart Garbage Monitoring and Cleaning System

Fig.3 in this system, cameras installed at specified locations on garbage bins to obtain pictures either at fixed time intervals or when enabled non- automatically through a control interconnection.

The acquired images are analysed using a deep learning driven artificial intelligence model that evaluates the fill percentage of the bin, detects overflow situations, and in certain cases classifies the category of waste. When the system identifies that a bin has reached capacity or is overflowing, it automatically generates a notification containing the bins location, visual data, and present condition, this notification is the. Transmitted to municipal officials and waste collection personnel via real time communication platforms such as telegram, facilitating prompt response and effective waste management.

Fig.3. Smart Dustbin Monitoring Process

The system maintains an electronic database of all observed waste bins, preserving historical information such as fill-level patterns, high-demand time frames, and the frequency of overflow incidents. Analyzing this data enables authorities to optimize waste pickup schedules, reduce operational expenses, and plan more effective collection pathways. Through the integration of AI-driven image processing, GPS tracking, and automatic notification alerts, the system ensures cleaner surroundings, minimizes health hazards resulting from unattended waste, and decreases the reliance on regular manual monitoring.

Furthermore, the software-centric architecture reduces dependency on physical components, leading to lower maintenance efforts and costs. The solution is highly scalable and can be conveniently deployed across multiple areas within smart city infrastructures. This proposed method demonstrates how artificial intelligence and automation can transform traditional urban waste management into a smarter, predictive, and highly efficient system.

B. RealVNC Viewer

RealVNC Viewer is a remote desktop application that allows users to observe and operate the AI-driven smart garbage monitoring and cleaning system from anywhere. It provides a secure connection to the Raspberry Pi or microcontroller via Wi-Fi or the internet, enabling real-time access to the system's interface. Through VNC Viewer, operators can remotely view live camera streams, AI-based detection outputs, sensor readings, and system records without being physically present at the site. This remote connectivity is highly useful for supervising waste levels, coordinating cleaning activities, analysing technical problems, and performing maintenance efficiently. Overall, RealVNC Viewer offers convenient control, operational flexibility, and effective management of the smart garbage monitoring system.

The smart garbage monitoring system employs RealVNC Viewer to provide live remote observation of dustbin status. When a bin reaches its capacity, the supervisor presses the Q key on the laptop, triggering the mounted camera to record the present condition of the bin. The system immediately transmits a "Bin Full" notification to Telegram, along with precise location information and captured images, enabling authorities to respond promptly.

Once the waste has been cleared, the sanitation worker confirms completion by pressing the push button positioned near the bin. This action automatically sends a "Bin Cleaned" message to Telegram with updated location data and images, securing accurate documentation. If necessary, the operator can press the Q key again after cleaning, allowing the system to verify the status and issue a "Bin Normal" alert, indicating the bin is ready for use.

Moreover, whenever the push button is pressed, the system sends a "Manual Button Alert" to Telegram regardless of the bin's status. This helps in safety and operational monitoring. By combining automated alerts with user validation, the system provides timely services, minimizing overflow issues,and provides a detailed digital monitoring and cleaning activities. The use of instant notifications, location tracking, and visual proof makes the system reliable, transparent, and well-suited for municipal waste management and smart city applications..

In the AI-driven smart garbage monitoring and cleaning system, RealVNC Viewer serves a key function by providing remote connectivity and control over the Raspberry Pi or other processing units linked to the garbage bins. RealVNC Viewer is a remote desktop tool that enables users to access and control the Raspberry Pi's display interface from a laptop, desktop, or mobile device via a network connection. This functionality is particularly valuable in smart waste management systems, where constant supervision of several bins located at different sites is necessary. Through RealVNC Viewer, the user can remotely watch live camera streams or review stored images captured by the cameras mounted on the bins.

When the AI component senses that a garbage bin has reached its capacity or is overflowing, the system automatically enables the user to access the associated images through RealVNC Viewer, allowing validation of the bin's condition without the need to be on site. This remote accessibility leads to quicker response actions and greatly minimizes the requirement for physical inspections, thereby reducing both time and workforce expenses. By integrating RealVNC Viewer with AI-driven image analysis and GPS-based alert systems, the solution establishes a seamless and effective operational flow: cameras capture images, the AI processes them to assess the bin status, and the results are presented via RealVNC Viewer for live observation and informed decision-making. This unified methodology enhances overall performance, encourages preventive waste management, and makes it easier to scale the system across multiple urban areas.

IV. MEASURING SETUP

A. Ai-Powered Smart Garbage Monitoring Fabrication

Fig4. implies that the Raspberry Pi take action as the main processing unit of the system, control all key operations such as image capture, AI-based analysis, and alert communication.

The Raspberry Pi is a compact, low-cost, and energy-efficient single-board computer that offers adequate processing capacity to run deep learning models for real-time waste recognition and grouping. Cameras are connected to the Raspberry Pi constantly capture real time images of the garbage bins by clicking the Q key, which are then examine locally using AI algorithms to point out the fill level and find overflow conditions. When a bin is found to be full or overflowing, the Raspberry Pi spontaneously generates alerts consists the bin's location and processed images are sends to municipal authorities or cleaning staff through messaging platforms such as Telegram.

Fig.4. Raspberry pi

In inclusion, the system supports remote access and monitoring through tools such as RealVNC Viewer, allowing operators to view live camera feeds, carry out maintenance tasks, and modernize the system without being physically available at the site. The flexibility, small size, and strong processing capacity of the Raspberry Pi build it appropriate for implementing a scalable and intelligent smart garbage keep track of solution. By incorporate real-time AI analysis, self-operating alert communication, and data-driven waste handling functions into a single programme, the system provide an well organized and practical perspective for latest urban environments.

Fig.5 implies that the significance of the GPS module in on condition that correct location instruction for each garbage bin. Moreover the system is basically based on image processing instead of physical level sensors, the GPS module make sure that every alert generated by the AI is linked to the command geographical position of the bin. When the camera captures images and the AI algorithm recognize a full or overflowing bin, the GPS module right away acquire the current complement and incorporate them in the alert message. This location particulars is then collective with municipal authorities or waste collection staff through platforms such as Telegram, authorize them to rapidly locate the bin that needs cleaning.

The GPS module in addition to helps diminish response time moreover supports greater arrangement of waste collection direction by clearly specify which bins are full and their expect locations. Over time, the location data collected through GPS allows dominance to study bin utilization trends in contrasting areas, recognize zones with high-rise waste generation, and determine where supplementary bins are needed. By integrating real-time location tracking with AI-based image inspection and automated alert systems, the GPS module enhance the accuracy, reliability, and overall efficiency of the smart garbage management system.

Fig.5.GPS Module

Fig. 6 shows the complete assembly of the AI-powered smart garbage bin, which collaborate intelligent software with necessary hardware components to design a compact and well organized system for automated waste monitoring and management. The Raspberry Pi acts as the central processing controller, synchronize all system functioning. A high-resolution camera is firmly secure to the bin to frequently detector the waste level or capture images when manually triggered. These images are sent to the Raspberry Pi, where a deep learning–based AI model inspects the bin condition to find out the fill level and recognize any overflow.

For accurate location identification, a GPS module is combined into the system so that the require coordinates of the bin are comprise in every alert. The whole system can be remotely retrieved and monitored using RealVNC Viewer, authorize operators to view live images, control system functions, and perform preservation without physically look in on the bin. All components are surround in a preservative housing mounted on the bin to secure durability, shielding from environmental conditions, and genuine cable organization.

When the AI discover that the bin is full, the system automatically coordinates with the captured image, GPS location, and bin current condition and sends an alert to municipal dominance or waste collection personnel through Telegram. This combined assembly results in a smart, self-monitoring garbage bin that utilize without physical level sensors, depends entirely on image-based intelligence, diminishes manual effort, enhance cleanliness, and improves the overall efficiency of waste collection operations.

Fig.6. Completed AI-powered smart garbage bin Assembly

V. Results

Fig.7 shows that the process used by the system to find when a garbage bin becomes full. The operator continuously observes the bin and presses the Q key to examine the current condition of the bin. When this Q key is pressed, the camera placed on the bin captures an image that indicate the level of waste inside. This captured image is used to verify that the bin has reached its maximal capacity. After verification, the system straight away sends a "Bin Full" alert to the Telegram channel along with the bin's GPS location. This knowledge allows the cleaning staff to the particular location which is comes along with the captured image, then the cleaning staff respond quickly, avoiding overflow and helping to keep a clean environment.

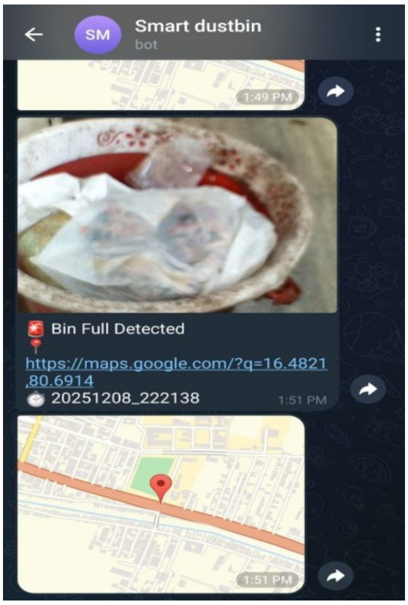

Fig.7. Bin Full Alert with GPS Notification

Fig. 8 shows that the cleaned garbage bin. If the waste is removed from the bin, the cleaning staff validate the completion of the task by pressing the manual push button placed near the garbage bin. This action implies a "Bin Cleaned" notification, which is sent to Telegram along with the updated location details. This confirmation make sure that the bin disinfect process has been decently completed and keeps all disturbed personnel informed. It also helps maintain correct records of disinfect activities, reduces the need for recurrent inspections, and make sure that the bin is ready for use afresh without any delay.

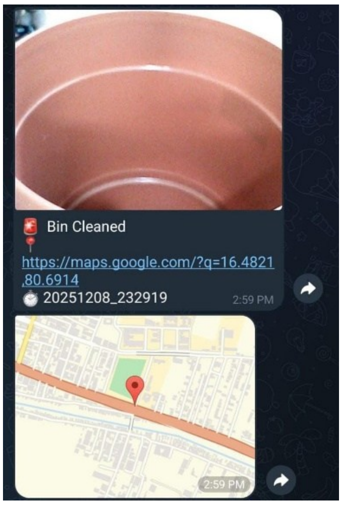

Fig.8.Bin Cleaned Status Update with GPS Notification

IV. Conclusion

Checking garbage overflow at the correct time is predominant for maintaining cleanliness in public and urban areas, and the amalgamation of camera-based monitoring with RealVNC Viewer provides an constructive solution. Rather than depending only on structured manual inspections, the system allows the operator to remotely give permission and detect dustbins through RealVNC Viewer from a single location. Whenever a bin needs to be inspect, the operator can activate image capture, and the system gives a clear visual affirmation of the bin's proper condition.

This cooperation between human direction using RealVNC Viewer and system-supported image conformation makes the observing process faster and simpler while diminish the chances of missed or delayed inspections. With the help of GPS tracking, bins spread over large areas can be easily recognize, authorize cleaning staff to come back quickly. Before choosing any action, the operator can checks the captured images, which helps keep away from false alerts. In inclusion, the saved images and location data can be used behind to study waste patterns and upgrade future planning and coordination.

Overall, this semi-automated system enhances cleanliness, reduces overflow incidents, improves operational efficiency, and supports effective municipal waste management.

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